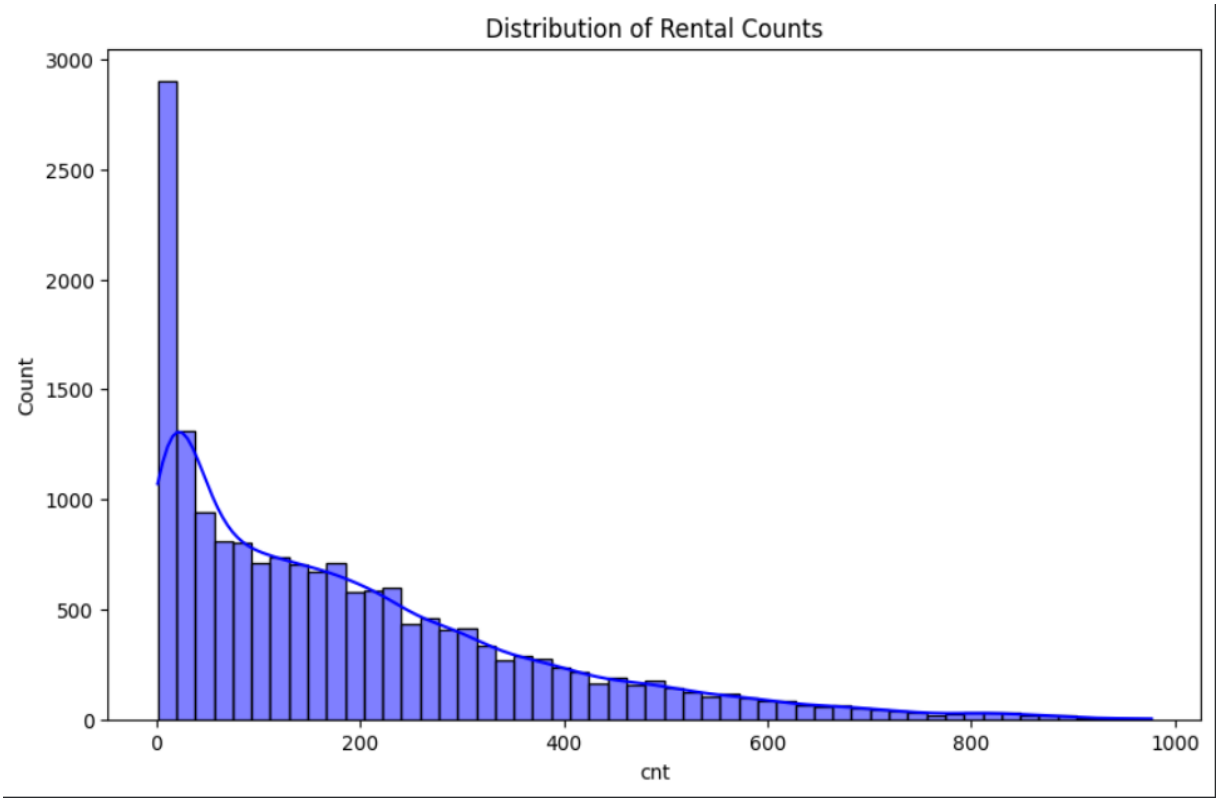
Project Execution : Bike Sharing Demand Forecasting

This report summarizes the end-to-end machine learning pipeline implemented to predict hourly bike rental demand.

Data Preprocessing: We started by loading the dataset, performing exploratory data analysis (EDA)



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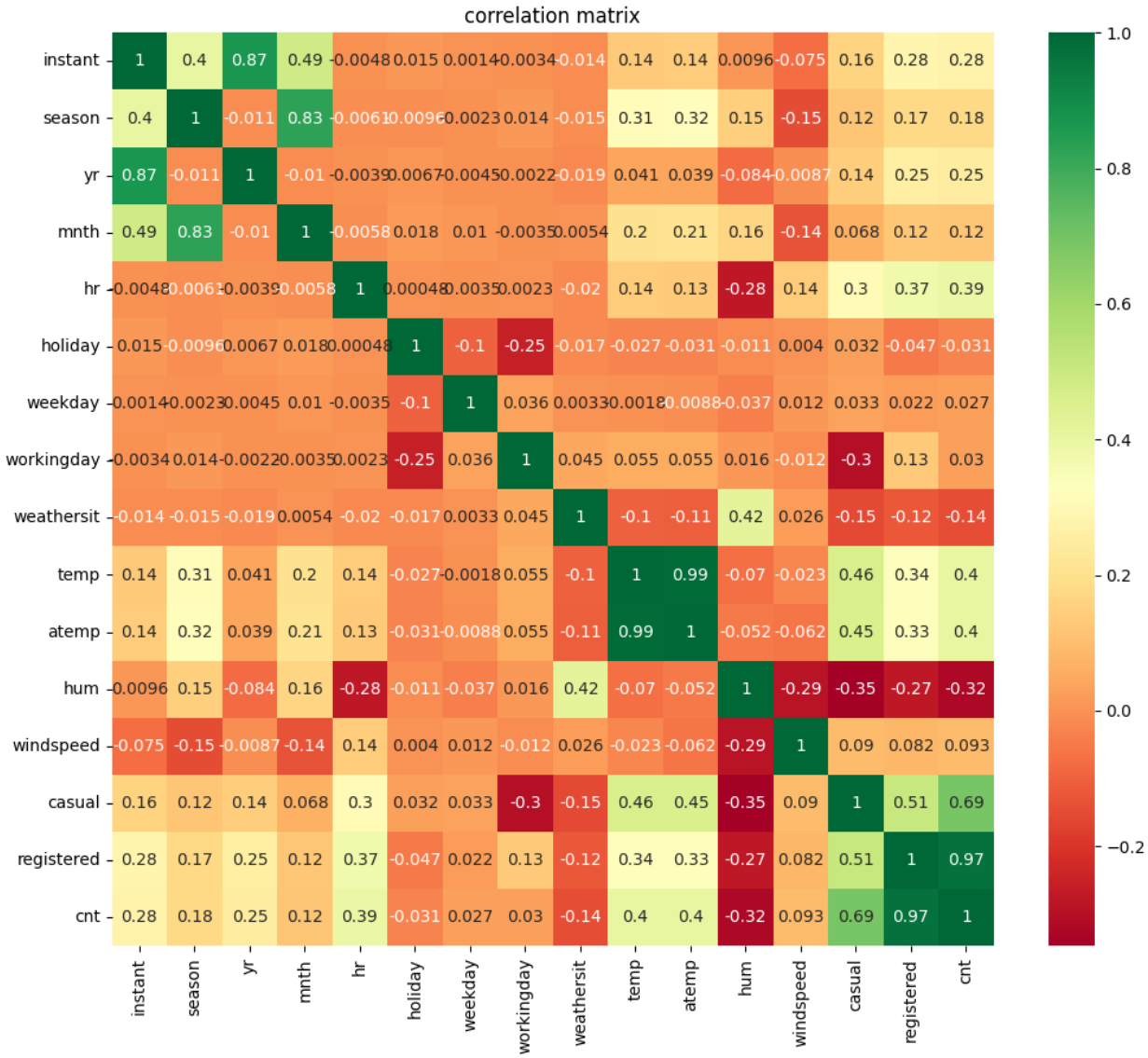
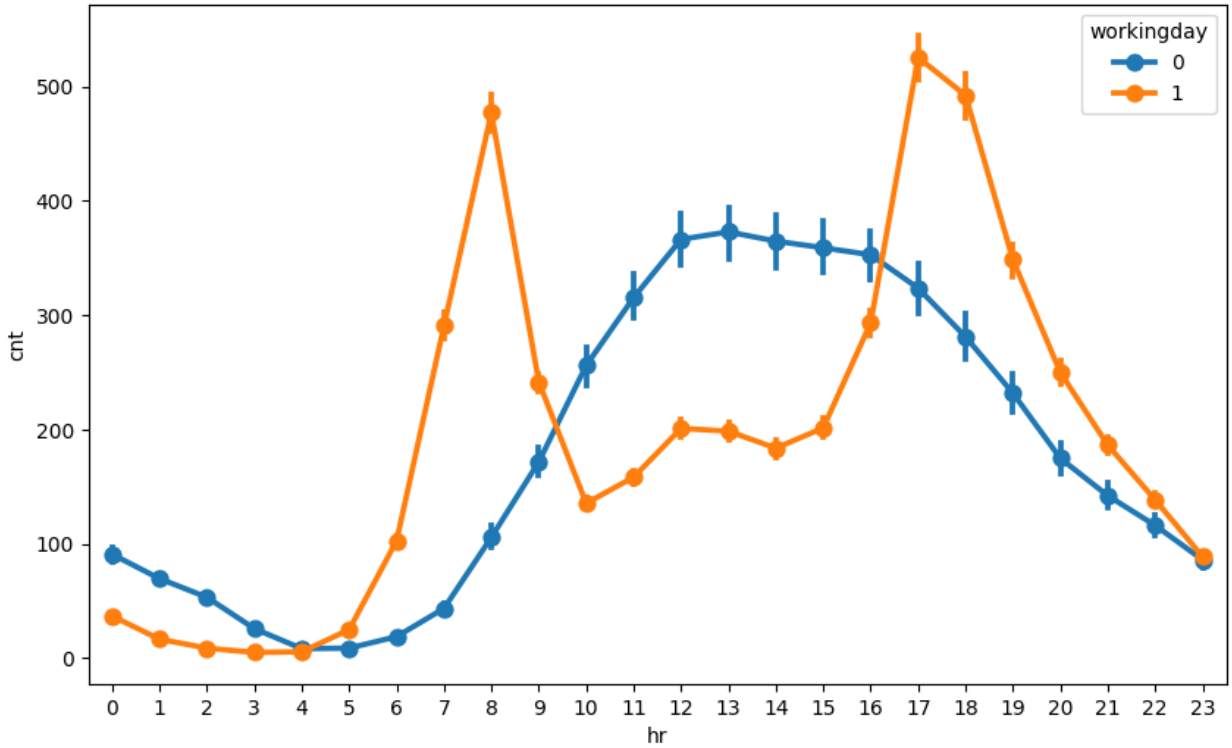

correlation matrix
instant 1 0.4 0.87 0.49 -0.0048 0.015 0.0014 -0.0034 -0.014 0.14 0.14 0.0096 -0.075 0.16 0.28 0.28
season 0.4 1 -0.011 0.83 -0.0061 0.0096 -0.0023 0.014 -0.015 0.31 0.32 0.15 -0.15 0.12 0.17 0.18
yr 0.87 -0.011 1 -0.01 -0.0039 0.0067 -0.0045 0.0022 -0.019 0.041 0.039 -0.084 -0.0087 0.14 0.25 0.25
mnth 0.49 0.83 -0.01 1 -0.0058 0.018 0.01 -0.0035 0.0054 0.2 0.21 0.16 -0.14 0.068 0.12 0.12
hr -0.0048 0.0061 -0.0039 -0.0058 1 0.00048 -0.0035 0.0023 -0.02 0.14 0.13 -0.28 0.14 0.3 0.37 0.39
holiday 0.015 -0.0096 0.0067 0.018 0.00048 1 -0.1 -0.25 -0.017 -0.027 -0.031 -0.011 0.004 0.032 -0.047 -0.031
weekday -0.0014 0.0023 0.0045 0.01 -0.0035 -0.1 1 0.036 0.0033 0.0018 0.0088 -0.037 0.012 0.033 0.022 0.027
workingday -0.0034 0.014 -0.0022 0.0035 0.0023 -0.25 0.036 1 0.045 0.055 0.055 0.016 -0.012 -0.3 0.13 0.03
weathersit -0.014 -0.015 -0.019 0.0054 -0.02 -0.017 0.0033 0.045 1 -0.1 -0.11 0.42 0.026 -0.15 -0.12 -0.14
temp 0.14 0.31 0.041 0.2 0.14 -0.027 -0.0018 0.055 -0.1 1 0.99 -0.07 -0.023 0.46 0.34 0.4
atemp 0.14 0.32 0.039 0.21 0.13 -0.031 -0.0088 0.055 -0.11 0.99 1 -0.052 -0.062 0.45 0.33 0.4
hum -0.0096 0.15 -0.084 0.16 -0.28 -0.011 -0.037 0.016 0.42 -0.07 -0.052 1 -0.29 -0.35 -0.27 -0.32
windspeed -0.075 -0.15 -0.0087 -0.14 0.14 0.004 0.012 -0.012 0.026 -0.023 -0.062 -0.29 1 0.09 0.082 0.093
casual 0.16 0.12 0.14 0.068 0.3 0.032 0.033 -0.3 -0.15 0.46 0.45 -0.35 0.09 1 0.51 0.69
registered 0.28 0.17 0.25 0.12 0.37 -0.047 0.022 0.13 -0.12 0.34 0.33 -0.27 0.082 0.51 1 0.97
cnt 0.28 0.18 0.25 0.12 0.39 -0.031 0.027 0.03 -0.14 0.4 0.4 -0.32 0.093 0.69 0.97 1
instant season yr mnth hr holiday weekday workingday weathersit temp atemp hum windspeed casual registered cnt
1.0
0.8
0.6
0.4
0.2
0.0
−0.2



and cleaning the data by handling outliers and removing redundant features (such as atemp due to high correlation with temp, ensuring no data leakage occurred).

Model Selection: We evaluated four regression models to identify the most suitable approach:

1 - Decision Tree Regressor

2 - Random Forest Regressor

3 - Gradient Boosting Regressor

4 - XGBoost Regressor

Hyperparameter Tuning: To optimize performance, we utilized GridSearchCV with cross-validation (cv=5) for each model, systematically testing various combinations of max_depth, n_estimators, and learning_rate to ensure high predictive accuracy.

Evaluation & Comparison: After training, we compared the models based on the Test R^2 Score. The Gradient Boosting Regressor was identified as the final production model due to its superior balance of accuracy (0.9597) and generalization capability.

Residual Analysis: We concluded the evaluation by plotting residuals to verify that the model's errors were random and that no systematic bias existed in the predictions.

Directed & Engineered by :

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